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Mixed practice. Integrate the following functions.

1) $\int e^{xe^x}\left(e^x + xe^x\right)\,dx$

2) $\int \dfrac{3x^2 - 2}{x^2 - 2x - 8}\,dx$

3) $\int x\sin^{-1}(x)\,dx$

Improper Integrals. Determine whether the following integral converges of diverges. If it converges, determine the value it converges to.

4) $\displaystyle\int_1^\infty \frac{\ln(x)}{x}\,dx$

5) $\displaystyle\int_0^\infty \frac{3}{x^3}\,dx$

6) $\displaystyle\int_1^\infty \frac{3}{x^3}\,dx$

7) $\displaystyle\int_0^\infty e^{-x}\,dx$

Numeric Integration. For problems 8 through 10 let $f(x) = 10\sin(x^2)$. Report all approximations to 4 decimal places of precision.

8) Approximate $\displaystyle\int_0^{\sqrt{\pi}} f(x)\,dx$ using the **trapezoid method** with 6 partitions. Then find the maximum error of your approximation.

9) Approximate $\displaystyle\int_0^{\sqrt{\pi}} f(x)\,dx$ using the **midpoint method** with 6 partitions. Then find the maximum error of your approximation.

10) Approximate $\displaystyle\int_0^{\sqrt{\pi}} f(x)\,dx$ using **Simpson's method** with 6 partitions. Then find the maximum error of your approximation.

It should be noted that the function in question for problems 8 through 10 is called "elementary". That is to say it can be created using a (potentially infinite) combination of only addition, subtraction, multiplication, division, and composition. The integral of that function however is "non-elementary". This means we cannot find the integral through traditional means. We will see other ways to perform calculus on functions like this near the end of the course. For the sake of having something to compare to I can say the following.

$$\int_0^{\sqrt{\pi}} 10\sin(x^2)\,dx \approx 8.948315$$

I have determined this fact using advanced techniques not available to us now, but your approximations should be near this value. Remember, this value has nothing to do with the maximum error bounds, I am merely putting it here for your peace of mind.

1) $\int e^{xe^x}\left(e^x + xe^x\right)dx$ $\qquad f'(x)g(x) + f(x)g'(x)$

$u = xe^x$

$\frac{du}{dx} = (1 \cdot e^x) + (x \cdot e^x)$

$e^x + xe^x$

$du = e^x(1+x)dx$

$\int e^u \, du$

$e^u + C$

$e^{xe^x} + C$

2) $\int \frac{3x^2-2}{x^2-2x-8}\,dx$

$3(x^2 - \frac{2}{3})$

$x^2 - 2x - 8$

$\int \frac{3(x^2-\frac{2}{3})}{(x-4)(x+2)}\,dx$

$\frac{3x^2-2}{x^2-2x-8} = 3 + \frac{(3x^2-2) - 3(x^2-2x-8)}{x^2-2x-8}$

$3x^2 - 2 - (3x^2 - 6x - 24)$

$6x + 22$

$\frac{3x^2-2}{(x-4)(x+2)} = 3 + \frac{6x+22}{(x-4)(x+2)}$

$\frac{A}{x-4} + \frac{B}{x+2}$

$6x + 22 = A(x+2) + B(x-4)$

$x = 4$

$46 = 6A$

$A = \frac{46}{6} \quad \frac{23}{3}$

$x = -2$

$10 = -6B$

$B = -\frac{10}{6} = -\frac{5}{3}$

$\frac{6x+22}{x^2-2x-8} = \frac{23}{3(x-4)} - \frac{5}{3(x+2)}$

$\int (3 + \quad\quad)\,dx$

$3x + \frac{23}{3}\ln|x-4| - \frac{5}{3}\ln|x+2| + C$

3) $\int x\sin^{-1}(x)\,dx$ $\quad \int u\,dv = uv - \int v\,du$

$u = \sin^{-1}(x) \quad dv = x\,dx$

$du = \frac{1}{\sqrt{1-x^2}}\,dx \quad v = \frac{x^2}{2}$

$\int u\,dv = \sin^{-1}(x)\frac{x^2}{2} - \int \frac{x^2}{2}\cdot\frac{1}{\sqrt{1-x^2}}\,dx$

$\frac{x^2}{2}\sin^{-1}(x) - \frac{1}{2}\int \frac{x^2}{\sqrt{1-x^2}}\,dx$

$\int \frac{x^2}{\sqrt{1-x^2}}\,dx$

$\frac{-(1-x^2)+1}{\sqrt{1-x^2}}$

$-\sqrt{1-x^2} + \frac{1}{\sqrt{1-x^2}}$

$-\int\sqrt{1-x^2}\,dx + \int\frac{1}{\sqrt{1-x^2}}\,dx$

$-\left(\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}(x)\right) + \sin^{-1}(x)$

$-\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}(x)$

$\int x\sin^{-1}(x)\,dx = \frac{x^2}{2}\sin^{-1}(x) - \frac{1}{2}\left(-\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}(x)\right) + C$

$\frac{x^2}{2}\sin^{-1}(x) + \frac{x}{4}\sqrt{1-x^2} - \frac{1}{4}\sin^{-1}(x) + C$

4) $\int_1^\infty \frac{\ln(x)}{x}\,dx$

$\lim_{t\to\infty} \int_1^t \frac{\ln(x)}{x}\,dx$

$u = \ln(x)$

$du = \frac{1}{x}\,dx$

$\int_0^{\ln(t)} u\,du = \left[\frac{u^2}{2}\right]_0^{\ln(t)} \quad \frac{(\ln(t))^2}{2} - 0$

$\lim_{t\to\infty} \frac{(\ln t)^2}{2} = \infty$

diverges

5) $\int_0^\infty \frac{3}{x^2}\,dx$

$\lim_{x\to 0^+} \frac{3}{x^3} = \infty$

Diverges

6) $\int_1^\infty \frac{3}{x^3}\,dx$

$\lim_{x\to 1^+} \frac{3}{x^3} = 3$

$\int_1^\infty \frac{3}{x^3}\,dx = \lim_{t\to\infty} \int_1^t 3x^{-3}\,dx$

$\int 3x^{-3}\,dx = 3\left(\frac{x^{-2}}{-2}\right)$

$-\frac{3}{2x^2}$

$\left[-\frac{3}{2x^2}\right]_1^t = \left(\frac{-3}{2t^2}\right) - \left(\frac{-3}{2(1)^2}\right)$

$\lim_{t\to\infty} -\frac{3}{2t^t} + \frac{3}{2} = 0 + \frac{3}{2} = \frac{3}{2}$ Converges

7) $\int_0^\infty e^{-x}\,dx = \lim_{t\to\infty} \int_0^t e^{-x}\,dx$

$\left[-e^{-x}\right]_0^t = -e^{-t} - (-e^0)$

$-e^{-t} + 1$

$1 - \frac{1}{e^t}$

$\lim_{t\to\infty} \left(1 - \frac{1}{e^t}\right) = 1 - 0 = 1$ Converges

8) $\int_0^{\sqrt{\pi}} 10\sin(x^2)\,dx$ by trapezoids

$\Delta x = \frac{b-a}{n} = \frac{\sqrt{\pi}-0}{6} \quad \frac{\sqrt{\pi}}{6}$

$\frac{\sqrt{\pi}}{6} \cdot \frac{1}{2}\left(10\sin^{0}(\emptyset) + 2(10\sin^{1}(\tfrac{\pi}{36})) + 2(10\sin^{2}(\tfrac{4\pi}{36})) + 2(10\sin^{3}(\tfrac{9\pi}{36})) + \dots\right.$
$\left. 2(10\sin^{4}(\tfrac{16\pi}{36})) + 2(10\sin^{5}(\tfrac{25\pi}{36})) + (10\sin^{6}(\pi))\right)$

→ desmos for calculation

$\simeq 8.6857$

Last time I missed a decimal I never went back and finished things... it's 11:45 & time to get some shuteye.